SIR and SEIR Models for Pandemic Spread

BMI 500

Hypothesis

(a) Infections peak around day 35-40 (~1200 infected, ~800 healthy).
(b) Births cause repeated smaller waves.
(c) β matters more than γ since it controls how fast the disease spreads.

Was it supported?

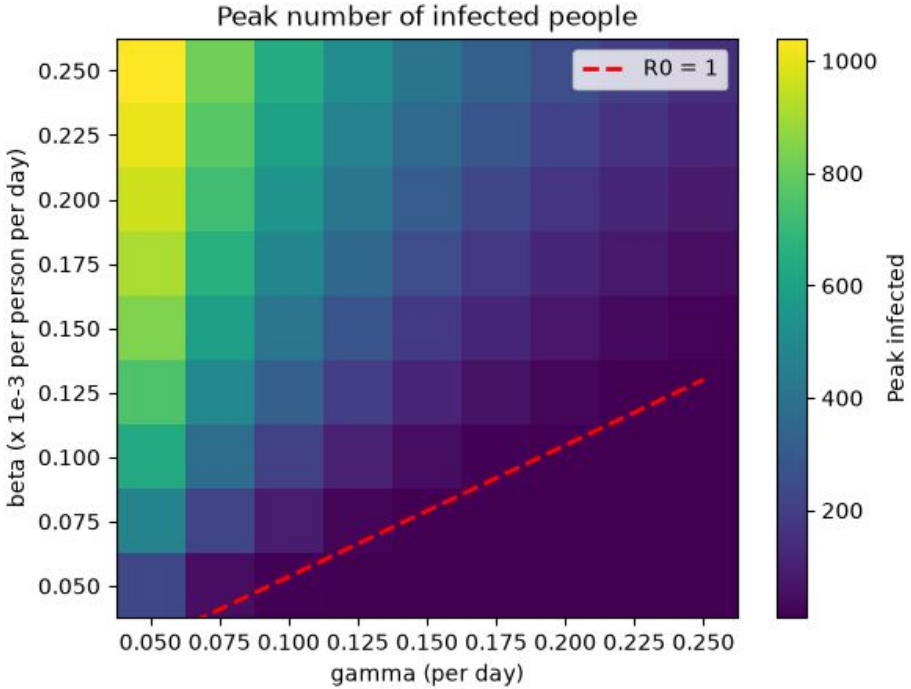
(a) Peak: day 43, and 800 (partly).
(b) Multiple Waves: supported.
(c) β vs γ: rejected, both had about the same effect and γ was slightly stronger.

A key number

R_0 = 2.5 (SIR) and 2.33 (SEIR) is the average number of people one sick person infects when everyone is susceptible. Both are above 1, so the disease spreads in either model; if R_0 were below 1, there would be no outbreak.

Lesson Learned

Adding births turned a single outbreak into repeating waves, but only because the lifespan was set to 200 days, so a model's structure and its parameter choices both shape the result.



Each square shows the highest number of people sick at once for one combination of spread rate (β, up) and recovery rate (γ, right). Brighter means a bigger outbreak. Faster spread and slower recovery (top-left) give up to about 1040 sick at once, while below the red line (R_0 = 1) the disease doesn't spread.